

C00_MPCSimulationLoop.m — program flow, objects and SIM -> LAB map

Line numbers refer to the delivered file. Preparation runs once; the main loop runs for every (initial condition, scenario) pair: $3 \times 4 = 12$ runs.

A. PREPARATION (once per execution)

Paths & output folder (L30-55)

```
addpath: models, flatness, configs, control/MPC,
        trajectories, utils, casadi
resultsBaseDir = <project>/results/MPC_C00/
fullRunDir = <yyyy-mm-dd>_Run_<NN> (first free NN)
```

SETTINGS (L57-89)

```
pos_threshold      = 0.05    % [m] in_track / t_conv
hover_tail_duration = 3.0    % [s]
perfect_state      = false
transmission_latency = 0.010 % [s] tx_time (placeholder)
rng_seed           = 42 ; make_gif = true
state_age_src = struct(mocap .040, px4 .020, esc .030) [s]
state_acquisition_delay = max(state_age_src) = 0.040 [s]
state_noise_std [16x1] (pos, vel, att, rates, T)
origin_xy struct: Circle, Lemniscate, Spiral, Lissajous
```

SYSTEM (L91-107)

```
mp = ModelParameters(); cp = ControlParameters()
tp = TrajectoryParameters(); Ts = cp.Ts
options = odeset('RelTol',1e-8,'AbsTol',1e-9) % plant
assert(state_acquisition_delay <= Ts)
model = C00_ANTX_quadcopter(mp,true)
mpc = C00_MPCController(model)
T_hover_pm = mp.m*mp.g/4 ; REFERENCE_OFFSET = 1
```

OSQP warm-up (L109-116)

```
x_warmup [16x1], xref_warmup [16x(p+2)], uref_warmup
mpc.init(x_warmup, xref_warmup, uref_warmup)
mpc.solve(x_warmup, xref_warmup(:,1:p+1), uref_warmup(:,1:p))
OSQP setup + first factorisation outside the timed loop
```

B. MAIN LOOP (L204-651) for c = 1: numel(initial_conditions, {OnReference, Perturbation, Origin}) for s = 1: size(scenarios,1)

B1 Scenario & reference (L213-224)

```
scenario_name T_end_base
x_ref_pwc [16x(N+1)]
u_ref_pwc [4x(N+1)]
run_dir = fullRunDir/<scenario>/<IC>
```

B2 Hover tail & used reference (L226-246)

```
x_hover_pt = x_ref_pwc(:,end); v=0; phi,theta=0;
            rates=0; T1..T4 = T_hover_pm
if Perturbation || Origin
    x_ref_use = [x_ref_pwc, repmat(x_hover_pt,1,n_tail)]
    u_ref_use = [u_ref_pwc, repmat(T_hover_pm*I(4),1,n_tail)]
    T_end_use = T_end_base + hover_tail_duration
else x_ref_use = x_ref_pwc ; T_end_use = T_end_base
x_hover = x_hover_pt ; u_hover = T_hover_pm*ones(4,1)
```

B3 Initial condition (L248-266)

```
current_x = x_ref_pwc(:,1) % OnReference
Perturbation: r_n += 0.20, r_e -= 0.20,
v = 0, phi=theta=psi = 0, p=q=r = 0
Origin: current_x = zeros(16,1);
current_x(1:2) = origin_xy.(scenario_name)
current_x(3) = -0.02 ; current_x(13:16) = cp.u_min_16
```

B4 Registers & mpc.init (L268-276)

```
u_now = u_ref_use(:,1) % OnReference
u_now = T_hover_pm*ones(4,1) % else
u_prev = u_now
mpc.init(current_x, ...
    x_ref_use(:,1+REFERENCE_OFFSET:end), ...
    u_ref_use(:,1+REFERENCE_OFFSET:end))
```

B5 Allocation (L278-303)

```
N_intervals = round(T_end_use/Ts) ; N_steps = N_intervals+1
N_intervals_base, N_steps_base (reference span only)
t_steps = (0:N_steps-1)*Ts
log_states [N_steps x 33] log_inputs [N_steps x 9]
log_errors [N_steps x 22] log_perst [N_steps x 16]
hist_t = 0 ; hist_x = current_x
sum_sq_pos = sum_sq_full = sum_sq_u = 0
solve_times, deadline_miss, qp_fail_log,
tx_times, total_latencies [N_intervals x 1]
```

B6 RECEDING HORIZON (L305-408)

```
for k = 1:N_intervals -> sheet 2
1 clock 2 acquisition 3 reference window
4 mpc.solve 5 status & latency 6 commit
7 plant [SIM] 8 registers 9 log 10 update
mpc.solve internals -> sheet 3
timing of one tick -> sheet 4
```

B7 Terminal sample (L410-442)

```
k_final = N_steps
e_final = current_x - ref_final
no MPC solve at t = T_end_use
log.*(k_final,:) completed
```

B8 Post-loop analysis (L444-502)

```
in_track_vec ; k_conv -> t_conv
rmse_ss_pos, rmse_ss_full (in-track samples only)
max_pos_err, max_yaw_err, max_att_err
avg/worst/std solve ; n_slow (> 0.8*Ts)
n_deadline_miss ; n_qp_fail ; avg_tx ; avg_total_latency
perstate_rmse ; rmse_pos_ref, rmse_full_ref ; console
```

B9 Figures (L504-611)

```
fig1 16 states, simulated vs reference (full duration)
fig2 applied inputs u applied vs u_ref, bounds
fig3 tracking analysis (reference span): position,
    attitude, yaw, thrust, per-state RMSE, solve time
saveas -> plot_states.png, plot_inputs.png,
    plot_analysis.png
```

B10 GIF & CSV (L613-630)

```
if make_gif: GenerateTrackingSimulationGIF(
    log_states, x_ref_use, gif, title, "2D", mp, C00_PlotDrone)
states.csv inputs.csv errors.csv per_state_errors.csv
```

B11 Metrics (L632-651)

```
metrics_summary {8 values}
metrics_extended {25 values}
```

B12 End of loops (L653-716)

```
MPC_metrics_summary.csv
MPC_metrics_extended.csv (25 cols)
Omega_i plausibility figure
local functions L718-765:
refWindow refAt delayedState
ternaryStateAge
```

next (c, s) pair (12 runs)

Reference generation (L118-131) [offline]

```
traj_circle = ShapeCircle(tp.Circle)
traj_lemniscate = ShapeLemniscate2(tp.Lemniscate)
traj_spiral = ShapeSpiral(tp.Spiral)
traj_lissajous = ShapeLissajous3D(tp.Lissajous3D)
[x_*, u_*, t_*, i_*] = C00_GeneratePWC_reference( ...
    traj, tp.T_duration_*, Ts, nlp_opts)
nlp_closed.periodic = true (Circle, Lemniscate, Lissajous)
nlp_open.periodic = false (Spiral); print_level = 0
x_ref [16x253], u_ref [4x253] (T = 25.2 s, N = 252)
```

Scenarios (L133-140)

```
scenarios{s,:} = {name, traj_obj, T_end, x_ref, u_ref}
s = 1 'Circle' 2 'Lemniscate' 3 'Spiral' 4 'Lissajous'
```

Reference validation (L142-153)

```
rep = C00_TrajectoryValidation(x_ref,u_ref,t,info,name,false)
if ~rep.pass -> warning 'RefFail' (the run continues)
```

Config JSON & log headers (L155-202)

```
simulation_config.json (model, control, timing, measurement)
state_headers (33) input_headers (9) error_headers (22)
state_labels input_labels angle_idx = [7 8 9]
```

Objects and who calls whom

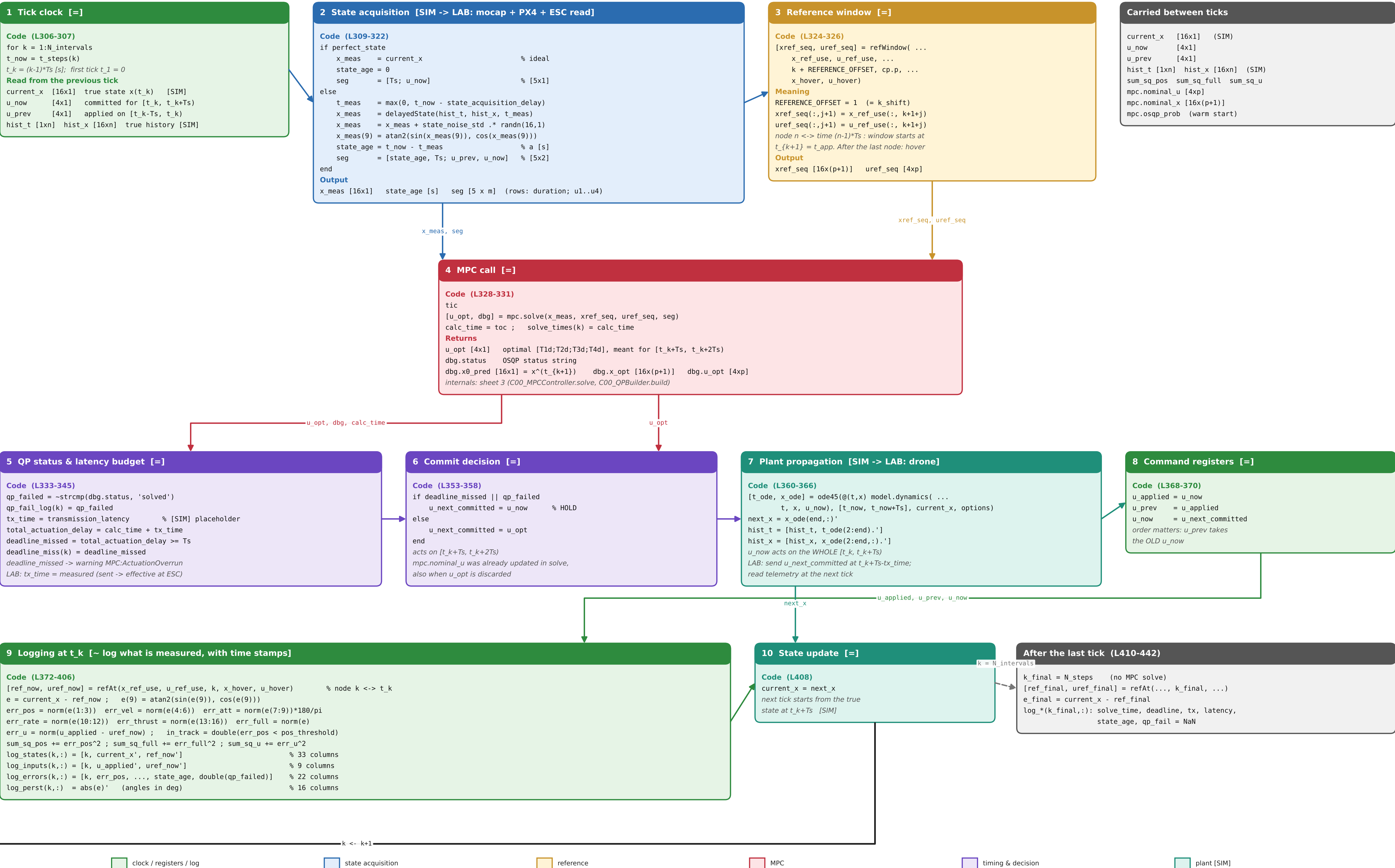
```
model = C00_ANTX_quadcopter(mp,true)
1) PLANT: model.dynamics in ode45 L362 [SIM]
2) PREDICTOR inside mpc: dynamics, get_jacobians
mpc = C00_MPCController(model)
mpc.init(...) L114, L276
mpc.solve(...) L115, L329
-> C00_QPBuilder.build, Discretization.discretize, osqp
C00_GeneratePWC_reference (offline)
-> C00_FlatnessMap, C00_ANTX_quadcopter, CasADi/IPOPT
C00_TrajectoryValidation (offline, ode45 replay)
GenerateTrackingSimulationGIF, C00_PlotDrone (plots)
```

SIM -> LAB replacement map

current_x (true state)	-> mocap + PX4 est. + ESC RPM
delayedState + noise	-> real age (timestamps) + noise
ode45 plant (L362)	-> drone driven through the GCS
transmission_latency	-> measured tx_time
calc_time	-> measured, same code
x_ref/u_ref (NLP in run)	-> loaded from the validated CSV
log of current_x	-> log of mocap / estimate
tick counter k	-> elapsed time since handover

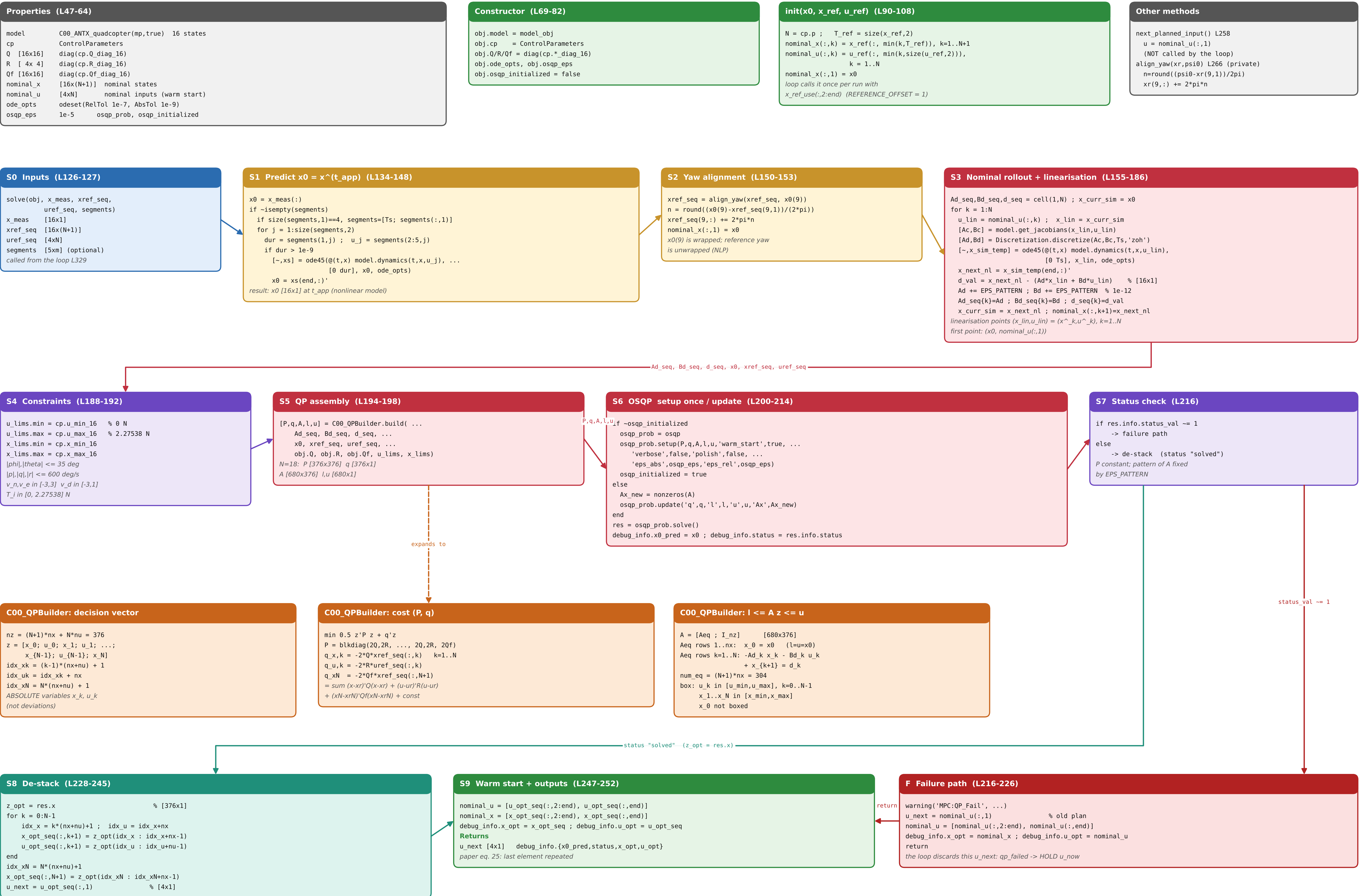
C00_MPCSimulationLoop.m — one receding-horizon tick k (lines L306-L408)

Tags: [SIM] = simulated element that is replaced in the lab | [=] stays identical in the lab. Line numbers refer to the delivered file.



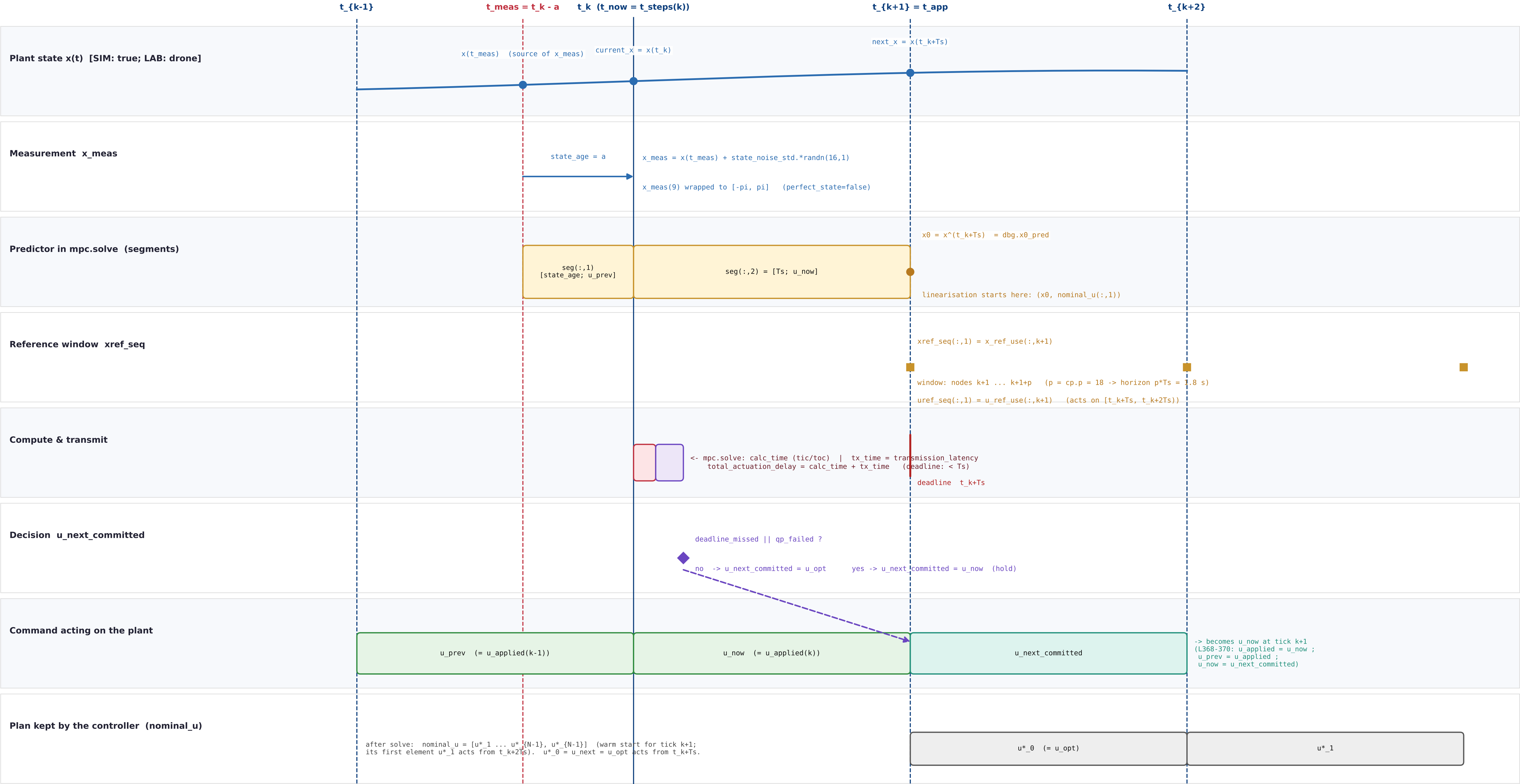
C00_MPCController.m + C00_QPBuilder.m — structure and mpc.solve(...) (lines L47-L260)

Notation: $N = cp.p = 18$ (horizon steps), $T_s = cp.T_s = 0.1$ s, $nx = 16$, $nu = 4$, $t_{app} = t_{\{k+1\}}$ (instant at which the returned input starts to act).



Timing of one tick k — C00_MPCSimulationLoop.m + C00_MPCController.solve (one-step actuation, history prediction)

Variables as in the code. $T_s = 0.1$ s. $a = \text{state_age} = t_{\text{now}} - t_{\text{meas}}$. $t_{\text{app}} = t_{\{k+1\}} = t_k + T_s$. Example drawn with $a = 0.040$ s, $\text{calc_time} = 0.008$ s, $\text{tx_time} = 0.010$ s (placeholders).



Tick $k+1$ repeats the same sequence one T_s later: its x_{meas} is $x(t_{\{k+1\}} - a)$, its history is $[\text{state_age}, T_s; u_{\text{prev}}, u_{\text{now}}]$ with $u_{\text{prev}} = (\text{old } u_{\text{now}})$ and $u_{\text{now}} = u_{\text{next_committed}}$.

The command computed at tick k is NOT applied before $t_k + T_s$: the plant integrates u_{now} over $[t_k, t_k + T_s]$. In the lab the send must be scheduled at $t_k + T_s - \text{tx_time}$ so that it becomes effective at t_{app} .

Variable dictionary — C00_MPCSimulationLoop.m and C00_MPCController.m / C00_QPBuilder.m

name | size | meaning. Sizes for cp.p = 18, nx = 16, nu = 4. [SIM] = exists only because the plant is simulated.

SimulationLoop: time, state, measurement			
name	size	meaning	
Ts		scalar	cp.Ts = 0.1 s sampling time
cp.p (N in controller)		scalar	horizon steps = 18
k		scalar	tick index 1..N_intervals
t_steps		1xN_steps	(0:N_steps-1)*Ts [s]
t_now		scalar	t_steps(k) = t_k [s]
N_intervals		scalar	round(T_end_use/Ts) = number of ticks
N_steps		scalar	N_intervals + 1 (samples logged)
current_x		16x1	TRUE plant state x(t_k) [SIM]
next_x		16x1	x(t_k+Ts) from ode45 [SIM]
hist_t		1xn	dense true-state time stamps [s] [SIM]
hist_x		16xn	dense true-state history [SIM]
x_meas		16x1	state given to mpc.solve
t_meas		scalar	max(0, t_now - state_acquisition_delay)
state_age		scalar	a = t_now - t_meas [s]
state_acquisition_delay		scalar	max(state_age_src) = 0.040 s
state_noise_std		16x1	noise sigma per state
REFERENCE_OFFSET		scalar	1 (= k_shift): window starts at node k+1
T_hover_pm		scalar	mp.m*mp.g/4 = 0.7137 N per motor

SimulationLoop: commands, timing, reference window			
name	size	meaning	
u_now		4x1	committed command, acts on [t_k, t_k+Ts)
u_prev		4x1	command applied on [t_k-Ts, t_k)
seg		5xm	[duration; u] history for the predictor
u_opt		4x1	mpc.solve output, for [t_k+Ts, t_k+2Ts)
u_next_committed		4x1	u_opt, or u_now (hold)
u_applied		4x1	= u_now, logged as applied input
dbg		struct	status, x0_pred, x_opt, u_opt
qp_failed		logical	~strcmp(dbg.status, 'solved')
calc_time		scalar	tic/toc around mpc.solve [s]
tx_time		scalar	transmission_latency [s] (placeholder)
total_actuation_delay		scalar	calc_time + tx_time [s]
deadline_missed		logical	total_actuation_delay >= Ts
x_ref_use, u_ref_use		16xT, 4xT	reference used (with hover tail)
x_hover, u_hover		16x1, 4x1	hover target beyond the last node
xref_seq		16x(p+1)	window: nodes k+1 .. k+1+p
uref_seq		4xp	window: nodes k+1 .. k+p
ref_now, uref_now		16x1, 4x1	reference at node k (t_k)
e		16x1	current_x - ref_now (yaw wrapped)

SimulationLoop: errors, logs, metrics			
name	size	meaning	
err_pos, err_vel		scalar	norm(e(1:3)), norm(e(4:6))
err_att		scalar	norm(e(7:9))*180/pi [deg]
err_rate, err_thrust		scalar	norm(e(10:12)), norm(e(13:16))
err_full, err_u		scalar	norm(e); norm(u_applied - uref_now)
in_track		0/1	err_pos < pos_threshold (0.05 m)
log_states		N_steps x 33	[k, state(16), ref(16)]
log_inputs		N_steps x 9	[k, u_applied(4), uref(4)]
log_errors		N_steps x 22	see error_headers (last: qp_fail)
log_perst		N_steps x 16	e per state (angles in deg)
solve_times, tx_times		N_intervals x 1	per-tick calc_time, tx_time
deadline_miss, qp_fail_log		N_intervals x 1	per-tick flags
sum_sq_pos/full/u		scalar	running sums for cumulative RMSE/MSE
t_conv, rmse_ss_pos		scalar	first 5 consecutive in_track; RMSE of in-track samples
n_qp_fail		scalar	unsolved QPs on the reference span
metrics_summary/_extended		cell	8 / 25 values per (IC, scenario)

Controller / QP: variables of mpc.solve			
name	size	meaning	
x0		16x1	x^(t_app): predicted initial state
segments		5xm	[duration; u]; a 4x1 u is read as [Ts; u]
dur, u_j		scalar, 4x1	one history segment (L141-143)
nominal_u (property)		4xN	warm-start inputs u^_k
nominal_x (property)		16x(N+1)	linearisation points x^_k
x_curr_sim		16x1	state along the nominal rollout
u_lin, x_lin		4x1, 16x1	linearisation point k
Ac, Bc		16x16, 16x4	continuous Jacobians (get_jacobians)
Ad, Bd		16x16, 16x4	exact ZOH discretisation (+1e-12)
x_next_n1		16x1	nonlinear one-step propagation (ode45)
d_val		16x1	x_next_n1 - (Ad*x_lin + Bd*u_lin)
Ad_seq, Bd_seq, d_seq		1xN cells	LTV model x+ = Ad x + Bd u + d
u_lims, x_lims		structs	.min/.max boxes from cp.*_16
P, q, A, l, u		see meaning	OSQP data: P 376x376, q 376x1, A 680x376, l,u 680x1
res		struct	OSQP result: .x, .info.status, .status_val
z_opt		376x1	stacked optimum [x_0;u_0;...;x_N]
x_opt_seq, u_opt_seq		16x(N+1), 4xN	de-stacked optimum
u_next, debug_info		4x1, struct	returned to the loop